LESSON 01 / 13

Integration reverses differentiation on an interval

A function F is an antiderivative of f on an interval I when $F' = f$ there. We write

$$\int f(x)\,dx = F(x) + C.$$

If $F_1' = F_2' = f$ on I, then $(F_1 - F_2)' = 0$. The Mean Value Theorem therefore gives $F_1 - F_2 = C$: all antiderivatives on that interval form one vertical family.

The differential dx identifies the variable. Thus $\int 2x\,dx = x^2 + C$, whereas $\int 2x\,dt = 2xt + C$ when x is independent of t.

WORKED EXAMPLE

Reverse the power, exponential and trigonometric rules

Linearity allows term-by-term integration:

$$\int x^3\,dx = \tfrac{x^4}{4} + C, \qquad \int x^5\,dx = \tfrac{x^6}{6} + C,$$

$$\int (3x^2 + 2x + 1)\,dx = x^3 + x^2 + x + C, \qquad \int (3x^2 + 2x)\,dx = x^3 + x^2 + C.$$

Reverse the chain rule and include the inner scale:

$$\int e^{2x}\,dx = \tfrac{1}{2}e^{2x} + C, \qquad \int e^{3x}\,dx = \tfrac{1}{3}e^{3x} + C,$$

$$\int \sin(2x)\,dx = -\tfrac{1}{2}\cos(2x) + C.$$

Every answer can be checked immediately by differentiating it.

The basic antiderivative library comes from derivatives

Integrand	Antiderivative	Condition
x^n	x^(n+1)/(n+1)	n is not -1
1 / x	ln|x|	x is nonzero
e^x	e^x	all real x
sin x	-cos x	all real x
cos x	sin x	all real x
sec^2 x	tan x	cos x is nonzero
csc^2 x	-cot x	sin x is nonzero
sec x tan x	sec x	cos x is nonzero

READ THE FIGURE
The exceptional power n=-1 produces a logarithm. Trigonometric formulas retain the domains on which their derivatives exist.

An initial value selects one member of the family

WORKED EXAMPLE
Use the condition after finding the general antiderivative
Suppose $F'(x) = 3x^2 + 2x + 1$ *and* $F(0) = 5$. *Integration gives*

$$F(x) = x^3 + x^2 + x + C.$$

Now $F(0) = C = 5$, *so the required function is*

$$F(x) = x^3 + x^2 + x + 5.$$

The initial condition does not change the derivative; it fixes the vertical shift.

COMMON TRAP
One constant per connected interval
On $(0, \infty)$, $\int x^{-1}dx = \ln x + C$; *on* $(-\infty, 0)$, *it is* $\ln(-x) + C$. *The compact formula* $\ln|x| + C$ *works on either interval separately. A single constant need not describe both branches simultaneously.*

LESSON 02 / 13

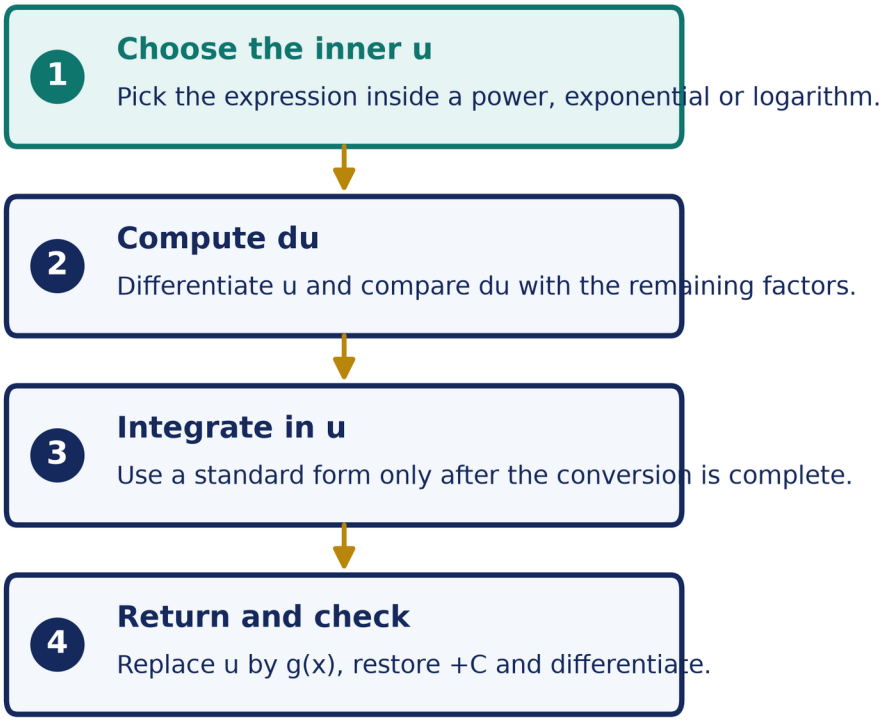
Look for an inner expression and its derivative

From $(F(g(x)))' = F'(g(x))g'(x)$, we obtain

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du, \qquad u = g(x), \quad du = g'(x)dx.$$

Substitution is not a guess followed by hope: the factor accompanying dx must account for du, up to a constant that can be moved outside the integral.

A substitution audit prevents a missing scale factor



READ THE FIGURE
A constant multiplier can be repaired. A missing nonconstant factor usually means that another method or another substitution is needed.

WORKED EXAMPLE

The derivative factor is already present

For

$$I = \int 2x(x^2+1)^5\,dx,$$

put $u = x^2+1$, so $du = 2x\,dx$. Then $I = \int u^5 du = (x^2+1)^6/6 + C$.

Likewise, with $u = x^2+3$,

$$\int 2x(x^2+3)^4\,dx = \frac{(x^2+3)^5}{5} + C.$$

WORKED EXAMPLE

A constant factor of one-half is enough

For $\int xe^{x^2}dx$, put $u = x^2$, so $du = 2x\,dx$ and $x\,dx = du/2$. Hence

$$\int xe^{x^2}dx = \tfrac{1}{2}\int e^u du = \tfrac{1}{2}e^{x^2} + C.$$

The same antiderivative is used in the source's definite questions P4 and Q7; their limits and numerical evaluation belong to Chapter 21.

Linear inner functions have a predictable scale

If $F' = f$ and $a \neq 0$, then

$$\int f(ax+b)\,dx = \tfrac{1}{a}F(ax+b) + C.$$

For example,

$$\int \cos(3x-1)dx = \tfrac{1}{3}\sin(3x-1) + C, \qquad \int (2x+1)^4 dx = \frac{(2x+1)^5}{10} + C.$$

COMMON TRAP

Do not invent the derivative factor

For $\int x(x^2+1)^5 dx$, the missing factor is only a constant 2, so substitution works with $x\,dx = du/2$. For $\int (x^2+1)^5 dx$, the needed factor $2x$ is absent and cannot be created as a constant multiplier.

LESSON 03 / 13

A numerator equal to the denominator's derivative gives a logarithm

Where $g(x) \neq 0$,

$$\int \frac{g'(x)}{g(x)}\,dx = \ln|g(x)| + C.$$

The absolute value is essential on negative branches. With $u = 3x + 2$,

$$\int \frac{dx}{3x+2} = \frac{1}{3}\ln|3x+2| + C.$$

Similarly,

$$\int \frac{2x+2}{x^2+2x+2}dx = \ln(x^2+2x+2) + C,$$

where the quadratic is always positive, so absolute-value signs are optional.

WORKED EXAMPLE

Tangent and cotangent are logarithmic substitutions

Since $\tan x = \sin x/\cos x$, put $u = \cos x$ on an interval where $\cos x \neq 0$:

$$\int \tan x\,dx = -\int \frac{du}{u} = -\ln|\cos x| + C = \ln|\sec x| + C.$$

In the same way,

$$\int \cot x\,dx = \ln|\sin x| + C.$$

Four standard trigonometric logs and their domains

Integral	Antiderivative	Valid away from
tan x	ln|sec x|	zeros of cos x
cot x	ln|sin x|	zeros of sin x
sec x	ln|sec x + tan x|	zeros of cos x
csc x	ln|csc x - cot x|	zeros of sin x

READ THE FIGURE

Each formula is proved by differentiation on one connected interval. Constants can change when a different interval is chosen.

Secant and cosecant use a multiplier that creates u'/u

For secant, let $u = \sec x + \tan x$. Then

$$\frac{du}{dx} = \sec x\tan x + \sec^2 x = \sec x(\sec x + \tan x) = \sec x\, u.$$

Therefore $u'/u = \sec x$, giving the table's formula. For cosecant, $u = \csc x - \cot x$ satisfies $u' = \csc x\, u$.

WORKED EXAMPLE

Check a scaled quotient rather than memorising its coefficient

$$\int \frac{6x}{3x^2+5}dx = \ln(3x^2+5) + C,$$

because the numerator is exactly the derivative of the denominator. In contrast,

$$\int \frac{x}{3x^2+5}dx = \frac{1}{6}\ln(3x^2+5) + C.$$

COMMON TRAP

A logarithm does not remove a singularity

The formulas above are valid only between consecutive excluded points. Never integrate through a zero of the original denominator as though $\ln|g(x)|$ made the integrand finite there.

LESSON 04 / 13

Simplify before choosing a technique

Many integrals become elementary after division, expansion or a trigonometric identity. Preparation is part of the calculation, not an optional cosmetic step:

$$\int \frac{x+1}{\sqrt{x}}dx = \int (x^{1/2} + x^{-1/2})dx = \frac{2}{3}x^{3/2} + 2x^{1/2} + C \qquad (x > 0).$$

WORKED EXAMPLE

A direct power substitution

For $\int \sin^3 x \cos x\, dx$, *put* $u = \sin x$, $du = \cos x\, dx$:

$$\int \sin^3 x \cos x\, dx = \int u^3 du = \frac{\sin^4 x}{4} + C.$$

Odd and even trigonometric powers need different preparation

For an odd power of sine, reserve one sine factor for $du = -\sin x\, dx$ *and convert the remaining even power with* $\sin^2 x = 1 - \cos^2 x$:

$$\int \sin^3 x\, dx = \int (1 - \cos^2 x)\sin x\, dx = -\cos x + \frac{\cos^3 x}{3} + C.$$

For even powers, halve the angle:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Consequently,

$$\int \sin^2 x\, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C, \qquad \int \cos^2 x\, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C.$$

These antiderivatives also underpin the source's definite questions P10, Q24 and Q36; evaluation over limits is deferred to Chapter 21.

Choose a trigonometric preparation from the powers

Pattern	Preparation	Likely next step
sin^n x cos x	u = sin x	power rule
cos^n x sin x	u = cos x	power rule
odd sine power	save sin x dx	u = cos x
odd cosine power	save cos x dx	u = sin x
even powers	half-angle identities	linear trigonometric terms
product sin x cos x	u = sin x or double angle	equivalent answers

READ THE FIGURE

The aim is to create either a complete du or a sum of standard forms. Do not mix the odd-power and even-power strategies midway.

WORKED EXAMPLE

Different-looking answers can all be correct

Three routes to $I = \int \sin x \cos x \, dx$ *give*

$$I = \tfrac{1}{2}\sin^2 x + C, \qquad I = -\tfrac{1}{2}\cos^2 x + C, \qquad I = -\tfrac{1}{4}\cos 2x + C.$$

Their derivatives are all $\sin x \cos x$*. Moreover,*

$$-\tfrac{1}{2}\cos^2 x = \tfrac{1}{2}\sin^2 x - \tfrac{1}{2}, \qquad -\tfrac{1}{4}\cos 2x = \tfrac{1}{2}\sin^2 x - \tfrac{1}{4}.$$

The apparent discrepancies are constants, not errors.

COMMON TRAP

Keep the original domain after cancellation

Algebraic simplification may expose a removable factor, but it does not restore points where the original integrand was undefined. State the simplified antiderivative only on intervals of the original domain.

LESSON 05 / 13

Recognise a sum of squares or a square-root difference

For $a>0$, direct differentiation gives

$$\int \frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+C, \qquad \int \frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}+C,$$

the second on $-a<x<a$. Thus

$$\int \frac{dx}{x^2+4}=\frac{1}{2}\tan^{-1}\frac{x}{2}+C, \qquad \int \frac{dx}{\sqrt{4-x^2}}=\sin^{-1}\frac{x}{2}+C.$$

The source's definite Q10–Q12 use these same antiderivatives before their limits are applied in Chapter 21.

WORKED EXAMPLE

Complete the square before using arctangent

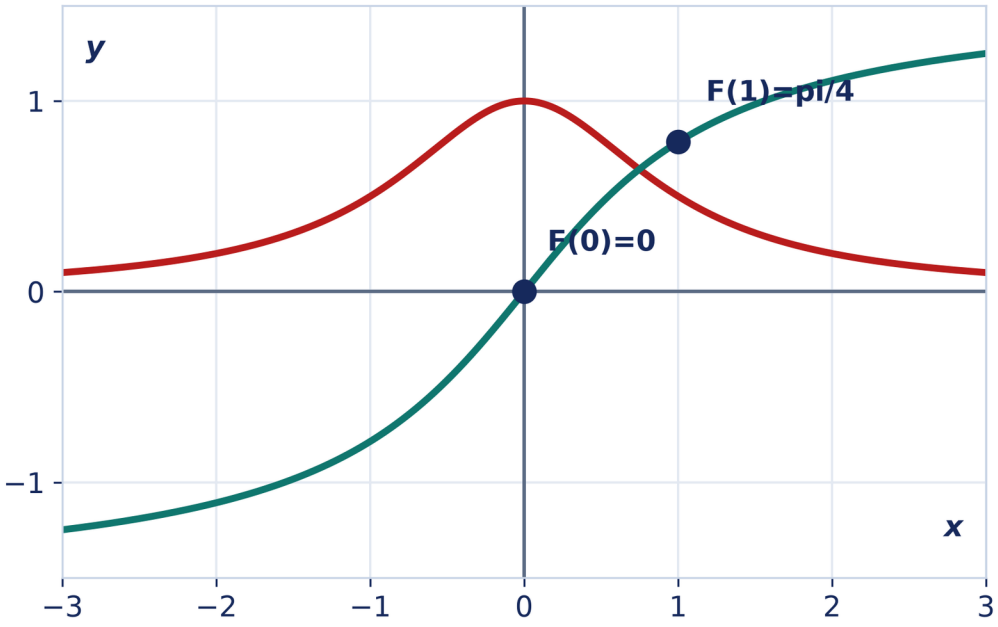
For SS,

$$x^2+2x+2=(x+1)^2+1.$$

With $u=x+1$ and $du=dx$,

$$\int \frac{dx}{x^2+2x+2}=\int \frac{du}{u^2+1}=\tan^{-1}(x+1)+C.$$

Arctangent has derivative 1/(1+x^2)



READ THE FIGURE

The teal curve is F(x)=arctan(x); the red curve is f(x)=1/(1+x^2). The slope of F at x=0 is 1 and at x=1 is 1/2: red gives height, teal gives the antiderivative.

Scaling must be handled before quoting the formula

WORKED EXAMPLE

Extract the coefficient of x squared

$$\int \frac{dx}{4x^2+9} = \frac{1}{4}\int \frac{dx}{x^2+9/4} = \frac{1}{6}\tan^{-1}\frac{2x}{3} + C.$$

Differentiating the last expression returns $1/(4x^2+9)$*, including the factor* $1/6$*.*

COMMON TRAP

Inverse notation is not reciprocal notation

Here $\tan^{-1}u$ *and* $\sin^{-1}u$ *mean* $\arctan u$ *and* $\arcsin u$*, not* $1/\tan u$ *or* $1/\sin u$*. Also, the arcsin formula requires* $a>0$ *and* $|x|<a$*; it is not valid at an arbitrary complex or outside-domain input.*

LESSON 06 / 13

Logarithms and inverse trigonometry come from the denominator's shape

For $a > 0$, on an interval avoiding singularities,

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C,$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right| + C.$$

The first is Q21. Its partial-fraction derivation is

$$\frac{1}{x^2 - a^2} = \frac{1}{2a} \left(\frac{1}{x - a} - \frac{1}{x + a} \right).$$

Three radical denominators have logarithmic antiderivatives

Integrand	Antiderivative	Domain interval
1/sqrt(x^2+a^2)	ln(x+sqrt(x^2+a^2))	all real x
1/sqrt(x^2-a^2)	ln|x+sqrt(x^2-a^2)|	x>a or x<-a
1/sqrt(a^2-x^2)	arcsin(x/a)	-a<x<a

READ THE FIGURE
The first two formulas can be checked by differentiating the radical and simplifying. On a disconnected domain, treat each interval separately.

A radical in the numerator may need a trigonometric substitution

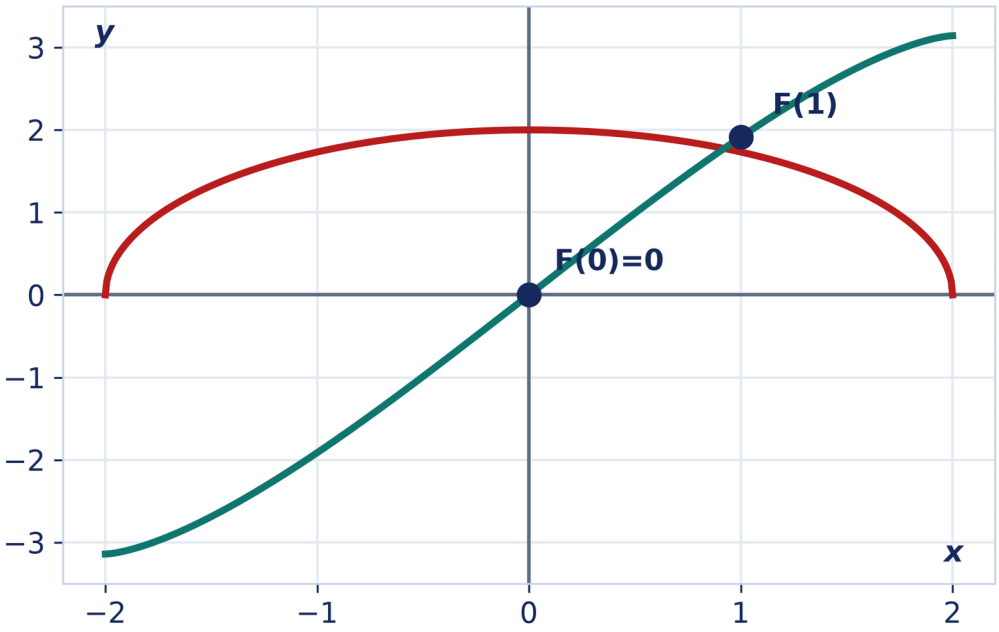
For $-a < x < a$, put $x = a \sin\theta$ with $-\pi/2 < \theta < \pi/2$. Then $dx = a\cos\theta\, d\theta$ and $\sqrt{a^2 - x^2} = a\cos\theta$, so

$$\int \sqrt{a^2 - x^2}\, dx = a^2 \int \cos^2\theta\, d\theta = \frac{a^2}{2}(\theta + \sin\theta\cos\theta) + C.$$

Returning to x gives

$$\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\frac{x}{a} + C.$$

The semicircle is the derivative of its antiderivative



READ THE FIGURE

The red semicircle is f(x)=sqrt(4-x^2); the teal curve is F=x sqrt(4-x^2)/2+2 arcsin(x/2). Differentiating the plotted F returns the red height.

COMMON TRAP

Do not confuse the two difference-of-squares orders

The answers for $1/(x^2 - a^2)$ and $1/(a^2 - x^2)$ differ by more than a casually moved sign. Their logarithmic ratios are different. Differentiation is faster and safer than trying to memorise the numerator order without a check.

LESSON 07 / 13

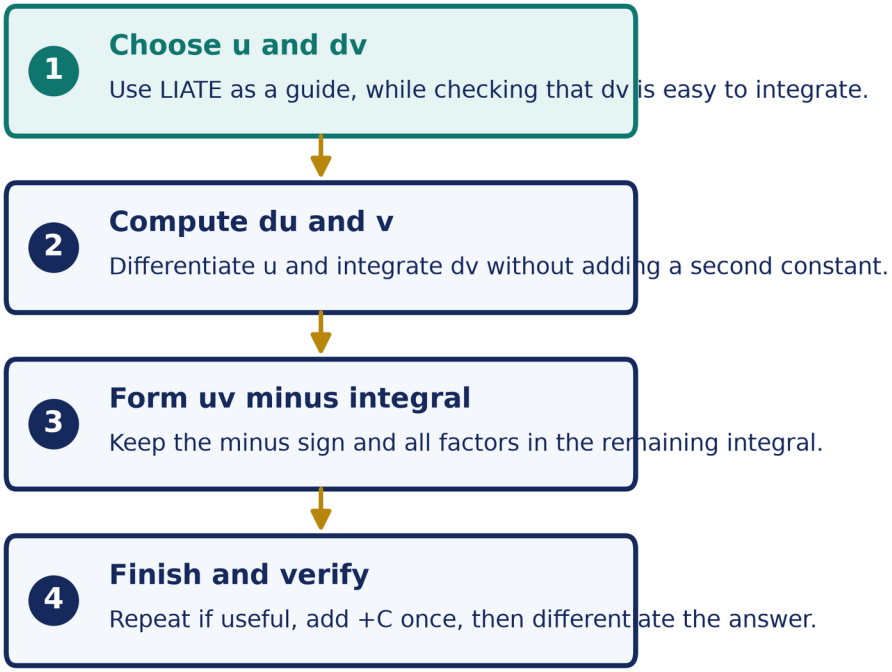
Integrate the product rule instead of guessing a product rule

From $(uv)' = u'v + uv'$, we obtain

$$\int u\,dv = uv - \int v\,du.$$

A successful choice makes $\int v\,du$ simpler than $\int u\,dv$. LIATE is a useful default: choose u from Logarithm, Inverse trigonometric, Algebraic, Trigonometric, Exponential, in that order.

By parts transfers difficulty from u to du



READ THE FIGURE
The method succeeds when du is simpler or the remaining integral has a known recurrence. Reversing a good choice often makes the problem harder.

WORKED EXAMPLE

One application removes the polynomial factor

For S6 and Q14, choose $u = x$ and $dv = e^x dx$. Then $du = dx$ and $v = e^x$:

$$\int xe^x dx = xe^x - \int e^x dx = e^x(x-1) + C.$$

WORKED EXAMPLE

A lone logarithm still has a product structure

For S7 and Q15 on $x > 0$, take $u = \ln x$ and $dv = dx$:

$$\int \ln x\,dx = x\ln x - \int x \cdot \tfrac{1}{x}\,dx = x\ln x - x + C.$$

Repeated parts can be organised as a polynomial recurrence

For P6 and Q16,

$$\int x^2e^x dx = x^2e^x - 2\int xe^x dx = x^2e^x - 2e^x(x-1) + C.$$

Therefore

$$\int x^2e^x dx = e^x(x^2 - 2x + 2) + C.$$

Differentiation gives $e^x(x^2 - 2x + 2) + e^x(2x - 2) = x^2e^x$.

WORKED EXAMPLE

Inverse trigonometric functions also benefit from dv=dx

$$\int \tan^{-1} x\,dx = x\tan^{-1} x - \int \frac{x}{1+x^2}dx = x\tan^{-1} x - \tfrac{1}{2}\ln(1+x^2) + C.$$

COMMON TRAP

Do not add independent constants at every by-parts step

A constant chosen for v can be absorbed into the remaining calculation. Write one final $+C$ for the complete antiderivative, and check that no intermediate constant has changed the intended recurrence.

LESSON 08 / 13

A product with sine may need only one application

For the antiderivative used in Q17,

$$\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x + C.$$

Its definite evaluation on $[0,\pi]$ is a Chapter 21 application of the Fundamental Theorem.

Exponential-trigonometric products can return to the original integral

WORKED EXAMPLE

Solve for the cyclic integral instead of abandoning it

Let $I = \int e^x \sin x\,dx$ and $J = \int e^x \cos x\,dx$. By parts,

$$I = e^x \sin x - J, \qquad J = e^x \cos x + I.$$

Substitute the second equation into the first:

$$I = e^x \sin x - e^x \cos x - I.$$

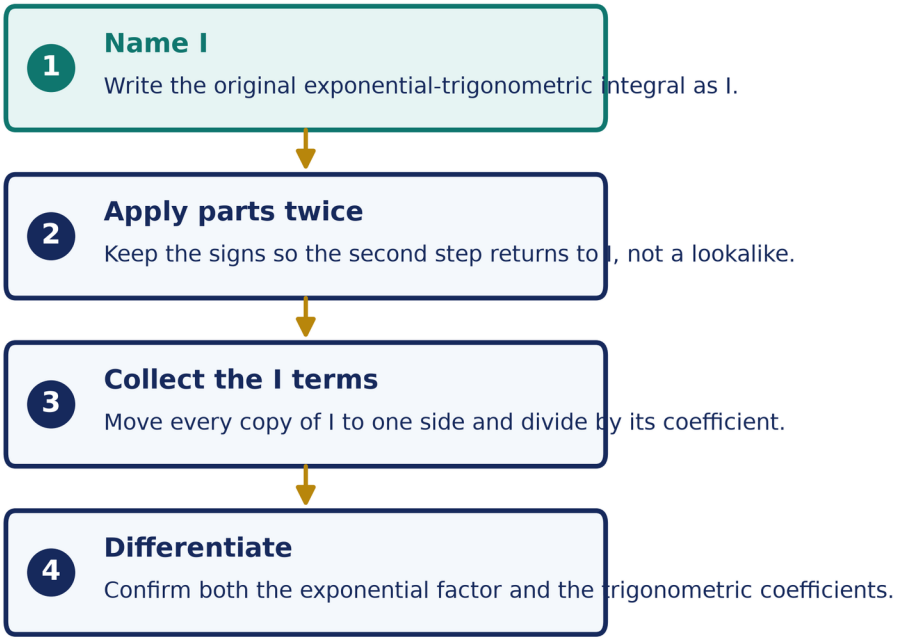
Thus P7 and Q18 have the antiderivative

$$\int e^x \sin x\,dx = \tfrac{1}{2}e^x(\sin x - \cos x) + C.$$

Differentiation gives

$$\tfrac{1}{2}e^x(\sin x - \cos x) + \tfrac{1}{2}e^x(\cos x + \sin x) = e^x \sin x.$$

A cyclic by-parts calculation ends with algebra



READ THE FIGURE
For e^x sin x, the calculation gives 2I=e^x(sin x-cos x). The factor 1/2 comes from collecting the returned integral.

An algebraic factor can hide a substitution after parts

WORKED EXAMPLE

Combine by parts with a logarithmic substitution

For $I=\int x\tan^{-1}x\,dx$, choose $u=\tan^{-1}x$ and $dv=x\,dx$:

$$I=\frac{x^2}{2}\tan^{-1}x-\frac{1}{2}\int\frac{x^2}{1+x^2}dx.$$

Since $x^2/(1+x^2)=1-1/(1+x^2)$,

$$I=\frac{x^2}{2}\tan^{-1}x-\frac{x}{2}+\frac{1}{2}\tan^{-1}x+C=\frac{x^2+1}{2}\tan^{-1}x-\frac{x}{2}+C.$$

COMMON TRAP

The returned integral must have exactly the same sign and coefficient

If two applications produce $I=\text{terms}-I$, then $2I=\text{terms}$. Writing $I=\text{terms}$ at that stage loses the essential factor $1/2$. If the result is $I=I$, no information has been gained; change the arrangement.

LESSON 09 / 13

First make the rational function proper

A rational function is proper when the numerator has smaller degree than the denominator. If not, divide first:

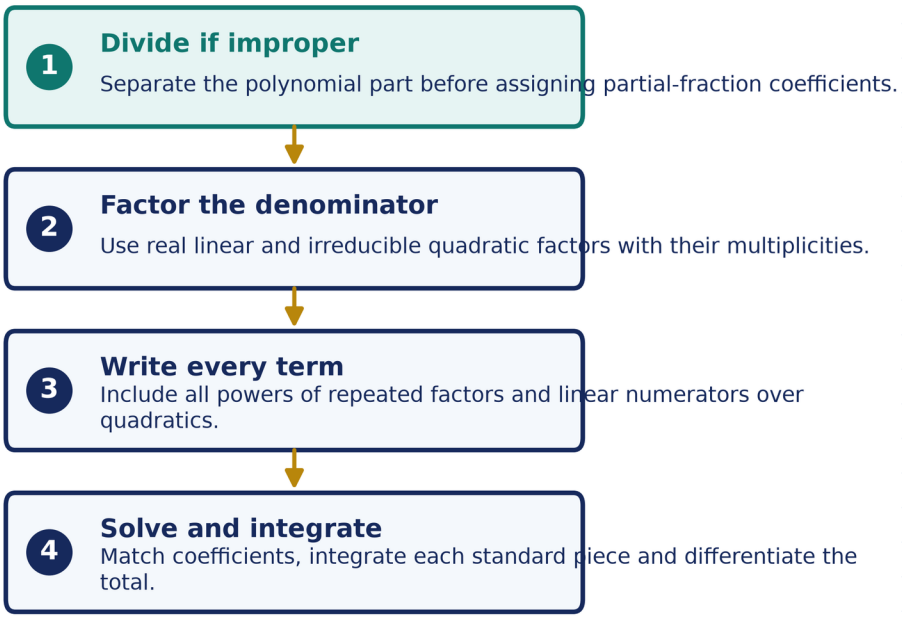
$$\frac{x^2+1}{x+1} = x - 1 + \frac{2}{x+1}.$$

Therefore

$$\int \frac{x^2+1}{x+1}dx = \frac{x^2}{2} - x + 2\ln|x+1| + C.$$

Only then factor the denominator and decompose the proper remainder.

A rational function has a reliable preparation order



READ THE FIGURE
Most partial-fraction errors occur before integration: no prior division, an omitted repeated power or an incomplete numerator over a quadratic.

Distinct linear factors produce logarithms

WORKED EXAMPLE

S8 and Q19: decompose before integrating

Since

$$\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2},$$

we obtain

$$\int \frac{dx}{(x+1)(x+2)} = \ln|x+1| - \ln|x+2| + C = \ln\left|\frac{x+1}{x+2}\right| + C.$$

The definite Q20 applies limits to this antiderivative in Chapter 21; no singularity may lie inside the interval used.

A repeated linear factor needs every power up to its multiplicity

For a factor $(x-a)^k$, include

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k}.$$

WORKED EXAMPLE

Repeated factor: one logarithm and one reciprocal power

Write

$$\frac{2x+3}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}.$$

Then $2x+3 = A(x+1) + B$, so $A = 2$ and $B = 1$. Hence

$$\int \frac{2x+3}{(x+1)^2}dx = 2\ln|x+1| - \frac{1}{x+1} + C.$$

COMMON TRAP

A single A/(x-a) is incomplete for a repeated factor

Omitting $(x-a)^{-2}, \ldots, (x-a)^{-k}$ usually makes the coefficient equations impossible—or, worse, produces an answer that matches at one point but has the wrong derivative. Differentiate the complete result on an allowed interval.

LESSON 10 / 13

Split the numerator into derivative plus constant

Let $D = x^2 + px + q$ be irreducible, so $D > 0$ after completing the square. For a numerator $Ax + B$, write

$$Ax + B = \alpha(2x + p) + \beta, \qquad \alpha = \frac{A}{2}, \quad \beta = B - \frac{Ap}{2}.$$

Then

$$\int \frac{Ax + B}{D} dx = \alpha \ln D + \beta \int \frac{dx}{D} + C.$$

The remaining integral becomes arctangent after completing the square.

Match the decomposition to the real denominator factor

Factor	Required term	Integrated shape
x-a	A/(x-a)	logarithm
(x-a)^k	terms through power k	log and reciprocal powers
irreducible quadratic	(Ax+B)/quadratic	log plus arctangent
improper rational	polynomial division first	polynomial plus proper part

READ THE FIGURE

The numerator over an irreducible quadratic must be linear. Splitting it into alpha D' + beta determines exactly when the logarithm term stands alone.

WORKED EXAMPLE

P8 correction: the remainder beta is zero

For

$$I = \int \frac{x+1}{x^2+2x+2}dx,$$

the denominator's derivative is $2x+2 = 2(x+1)$. *Thus*

$$x+1 = \tfrac{1}{2}(2x+2) + 0,$$

so the complete answer is

$$I = \tfrac{1}{2}\ln(x^2+2x+2) + C.$$

The source adds $\tfrac{1}{2}\tan^{-1}(x+1)$. *That extra term is incorrect: its derivative is*

$$\tfrac{1}{2}\cdot\frac{1}{(x+1)^2+1} = \frac{1}{2(x^2+2x+2)} \neq 0.$$

Differentiating the corrected answer returns exactly $(x+1)/(x^2+2x+2)$.

WORKED EXAMPLE

A nonzero remainder produces both standard shapes

For $D = x^2+2x+5 = (x+1)^2+4$,

$$3x+5 = \tfrac{3}{2}(2x+2) + 2.$$

Therefore

$$\int \frac{3x+5}{x^2+2x+5}dx = \tfrac{3}{2}\ln(x^2+2x+5) + 2\int \frac{dx}{(x+1)^2+4}.$$

The last integral is $\tfrac{1}{2}\tan^{-1}((x+1)/2)$, *hence*

$$I = \tfrac{3}{2}\ln(x^2+2x+5) + \tan^{-1}\tfrac{x+1}{2} + C.$$

COMMON TRAP

Check the coefficient split before integrating either piece

A quick derivative check of the proposed identity $Ax+B = \alpha D' + \beta$ *catches a missing remainder immediately. The logarithm and arctangent terms are not interchangeable decorations; their coefficients are forced by that identity.*

LESSON 11 / 13

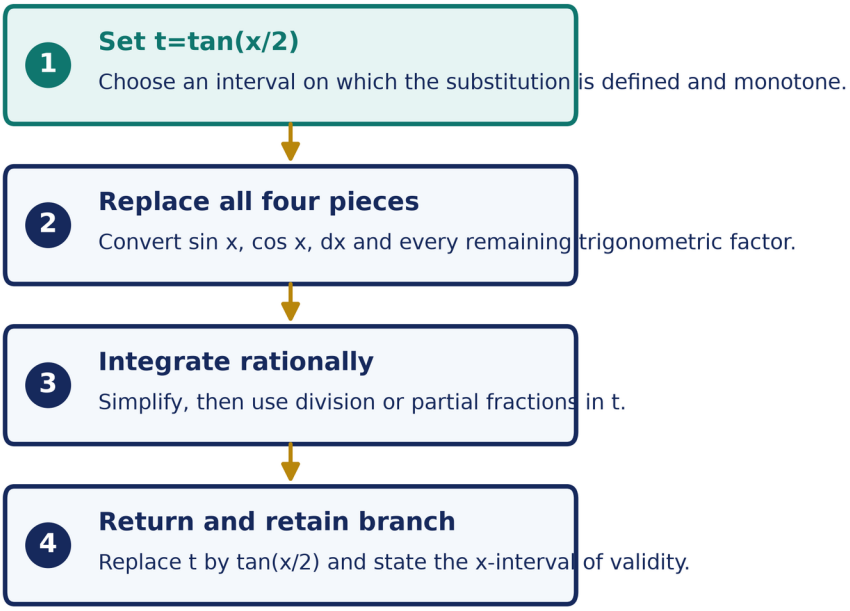
Convert sine and cosine into rational functions of one variable

For a rational expression in $\sin x$ and $\cos x$, the universal substitution is

$$t = \tan\frac{x}{2}, \qquad \sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad dx = \frac{2\,dt}{1+t^2}.$$

It is powerful but algebra-heavy, so first check for a simpler identity, symmetry or direct substitution.

The half-angle substitution is a complete change of variable



READ THE FIGURE

The factor dx=2dt/(1+t^2) is essential. Omitting it can coincidentally produce a familiar expression but cannot produce the correct antiderivative.

WORKED EXAMPLE

The antiderivative behind S13 and Q35

For

$$I=\int \frac{dx}{2+\cos x},$$

the denominator becomes

$$2+\cos x=2+\frac{1-t^2}{1+t^2}=\frac{3+t^2}{1+t^2}.$$

Including $dx=2dt/(1+t^2)$ gives

$$I=\int \frac{2}{3+t^2}dt=\frac{2}{\sqrt{3}}\tan^{-1}\frac{t}{\sqrt{3}}+C.$$

Thus, on $-\pi<x<\pi$,

$$I=\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{\tan(x/2)}{\sqrt{3}}\right)+C.$$

The definite value on $[0,\pi/2]$ is completed in Chapter 21.

Another route can give an equivalent-looking answer

For $\int dx/(1+\sin x)$, the same substitution gives

$$\int \frac{2\,dt}{(1+t)^2}=-\frac{2}{1+t}+C=-\frac{2}{1+\tan(x/2)}+C.$$

On an allowed branch this differs from $\tan x-\sec x$ only by a constant, and both differentiate to $1/(1+\sin x)$.

COMMON TRAP

The substitution has branches

$t=\tan(x/2)$ is undefined at $x=(2k+1)\pi$. The displayed arctangent antiderivative is not automatically one continuous antiderivative across those points. Use it on a single branch; definite problems also require an interval free of the original integrand's singularities.

LESSON 12 / 13

Diagnose the structure before calculating

The fastest method is usually the one exposed by preparation. Do not begin with a favourite technique and force the integrand to fit it.

A compact method-selection map

Visible structure	First method	Decisive check
polynomial or basic library	reverse derivative	power and scale
f(g(x)) g'(x)	u=g(x)	du is complete
g'(x)/g(x)	log substitution	g is nonzero
sum or difference of squares	standard form	complete square first
polynomial times e^x or trig	by parts	du gets simpler
log or inverse trig factor	by parts with dv=dx	new integral is easier
rational function	division and partial fractions	factor multiplicities
rational sin and cos	half-angle substitution	branch and dx factor

READ THE FIGURE
Preparation can move an integrand from one row to another. If a calculation grows without creating a standard form, stop and reassess the structure.

Three short choices show the difference

WORKED EXAMPLE

Substitution, parts and preparation are not interchangeable

For $\int xe^{x^2}dx$, the factor x helps create $du = 2x\,dx$:

$$\int xe^{x^2}dx = \tfrac{1}{2}e^{x^2} + C.$$

For $\int xe^{x}dx$, there is no inner x^2; by parts gives

$$\int xe^{x}dx = e^{x}(x-1) + C.$$

For $\int (x^{1/2} + x^{-1/2})dx$, no advanced method is needed:

$$\int (x^{1/2} + x^{-1/2})dx = \tfrac{2}{3}x^{3/2} + 2x^{1/2} + C.$$

Verification is a mathematical step, not a courtesy

A complete check has four parts:

1. Differentiate the proposed antiderivative and simplify to the **original** integrand.
2. Confirm the inner scale, signs and partial-fraction coefficients.
3. State an interval on which both integrand and antiderivative are defined.
4. If two answers look different, test whether their difference is constant on that interval.

For example, the three answers to $\int \sin x\cos x\,dx$ in `jee-04.md` differ by 0, $-1/2$ or $-1/4$ when written relative to $\tfrac{1}{2}\sin^2 x$. Their equality up to C is therefore proved, not inferred from similar graphs.

COMMON TRAP

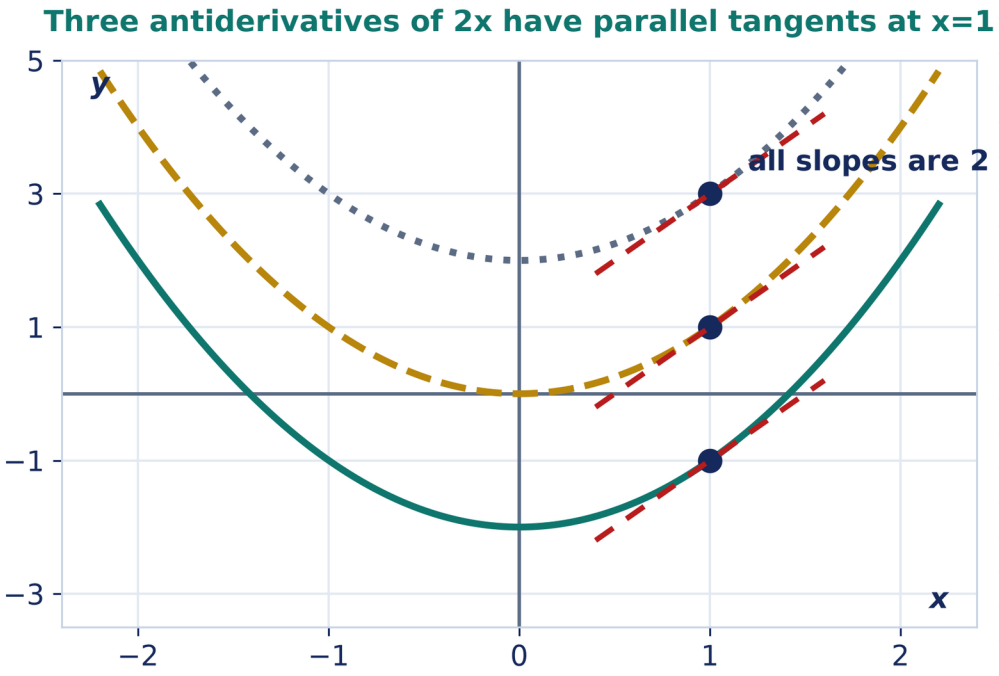
A value check at one point is not enough

Two different functions can agree at one point. Differentiation checks the whole relationship at once. Conversely, if $G' = f$ but the domain has been ignored, the calculation is still incomplete: singularities and disconnected branches remain part of the answer.

LESSON 13 / 13

An antiderivative is not merely an algebraic string. These six sheets display vertical families, domains, slopes, equivalent forms and substitution branches. Every curve is generated from an editable expression and independently differentiated in the verification suite.

Lab 1 — The constant moves a curve without changing its slopes



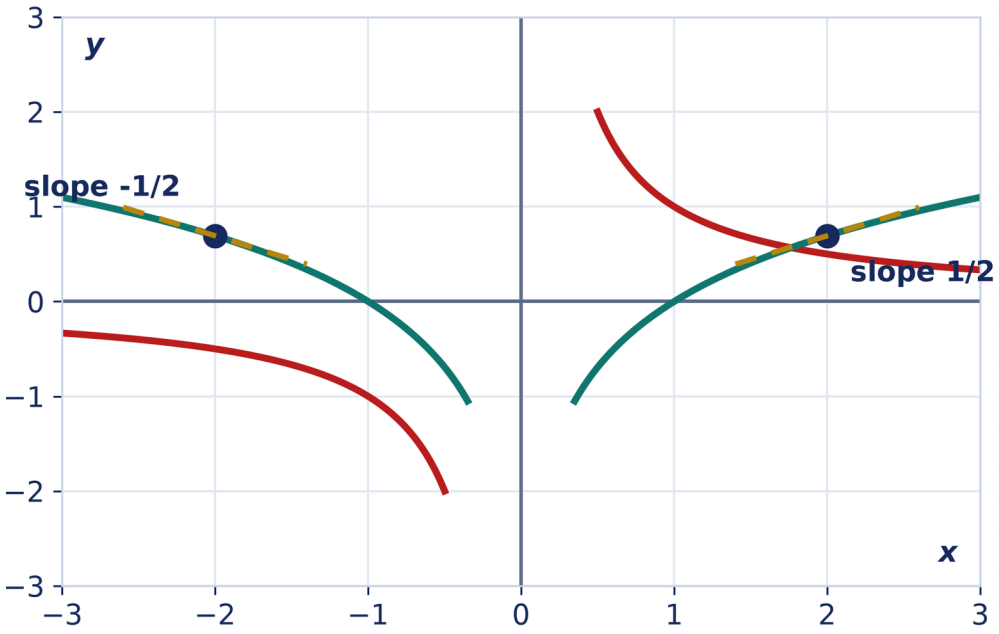
READ THE FIGURE

The curves x^2-2, x^2 and x^2+2 differ only by vertical shifts. Their red tangent segments at x=1 are parallel because each derivative is 2x=2.

If $F' = f$, then $(F + C)' = f$. Conversely, on one interval, any two antiderivatives have zero derivative for their difference and therefore differ by a constant. An initial condition selects exactly one of these vertical translates.

Lab 2 — Logarithm branches and their slopes

ln|x| differentiates to 1/x on both sides of zero



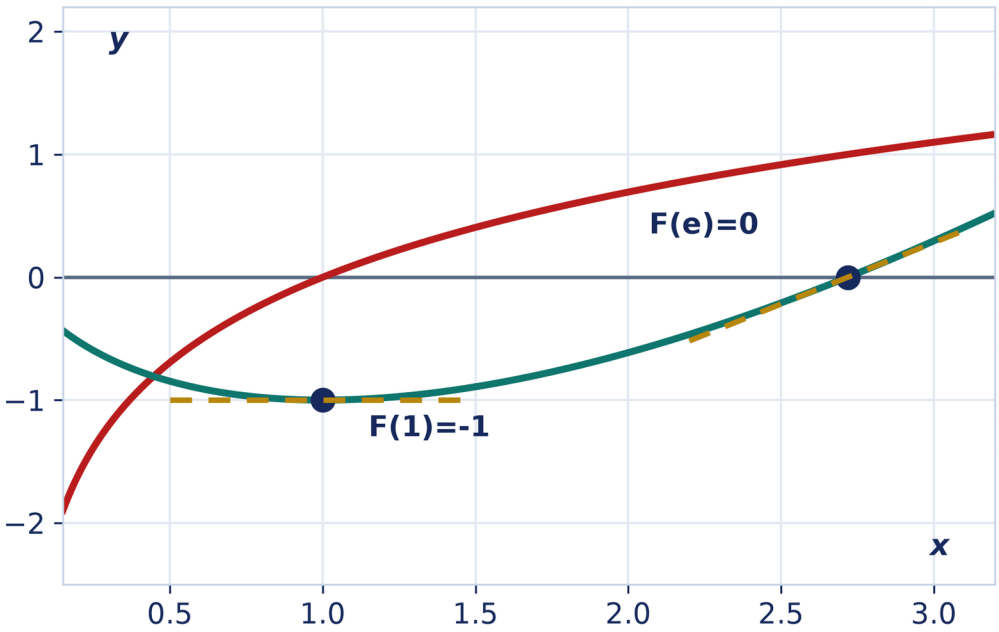
READ THE FIGURE

Red branches are f(x)=1/x; teal branches are F(x)=ln|x|. The gold tangent slopes at x=-2 and x=2 are -1/2 and 1/2. No curve crosses the excluded line x=0.

For $x<0$, $\ln|x| = \ln(-x)$, whose derivative is $(-1)/(-x) = 1/x$. The absolute value therefore handles both signs. The two disconnected branches may carry different constants even though the shorthand writes one C.

Lab 3 — By parts produces a function whose slope is ln x

The antiderivative x ln x - x has tangent slope ln x

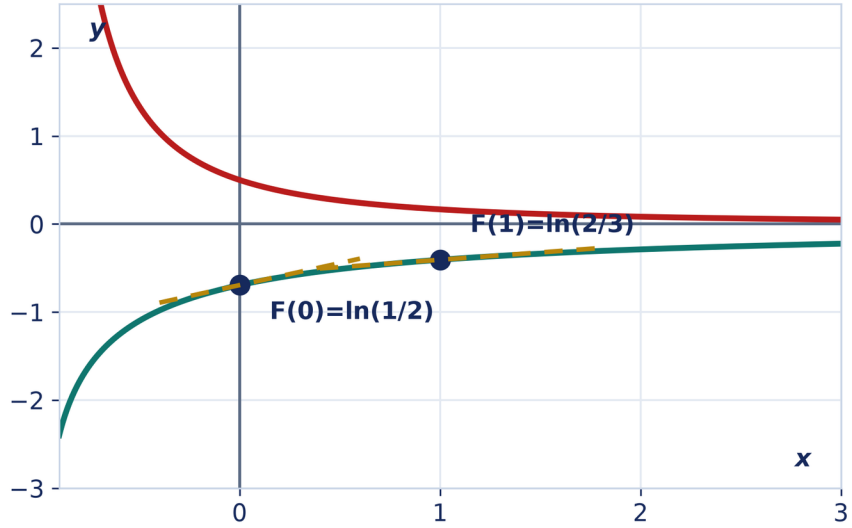


READ THE FIGURE
Red is f(x)=ln x and teal is F(x)=x ln x−x. The gold tangent at x=1 is horizontal because ln 1=0; at x=e its slope is 1 because ln e=1.

*The product-rule calculation $\int \ln x\,dx = x\ln x - \int 1\,dx$ is visible in the slopes. Notice that the red curve gives the **slope** of the teal curve, not its height. This distinction is the geometric meaning of $F' = f$.*

Lab 4 — Partial fractions make a logarithm increase

A positive partial-fraction integrand makes its antiderivative increase



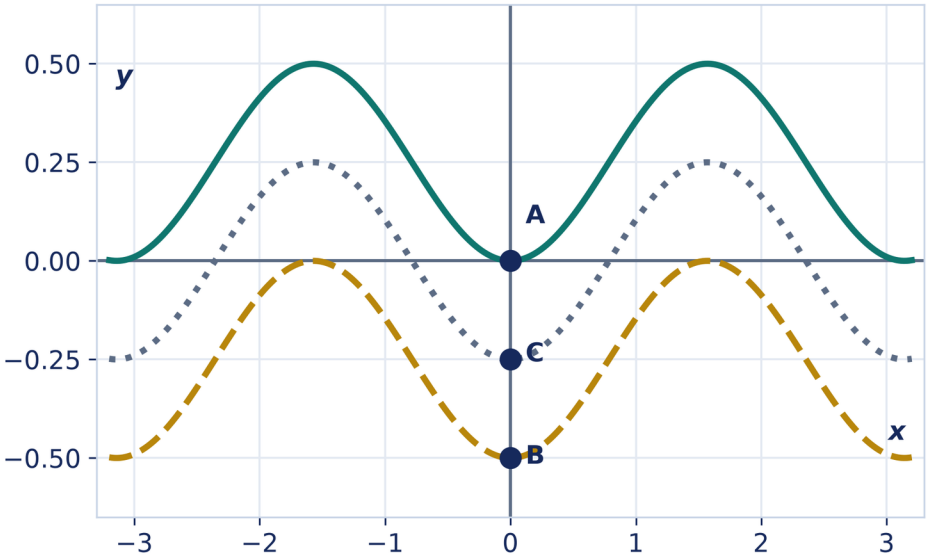
READ THE FIGURE

On x>-1, red is f=1/((x+1)(x+2)) and teal is F=ln|(x+1)/(x+2)|. Gold tangent slopes are f(0)=1/2 and f(1)=1/6, so F increases with decreasing slope.

The decomposition $f = 1/(x+1) - 1/(x+2)$ explains why the antiderivative is a difference of logarithms. The graph uses only the interval $x > -1$; the excluded points -2 and -1 divide the full domain into separate branches.

Lab 5 — Equivalent answers are vertical translates

Three correct antiderivatives of sin x cos x have the same shape

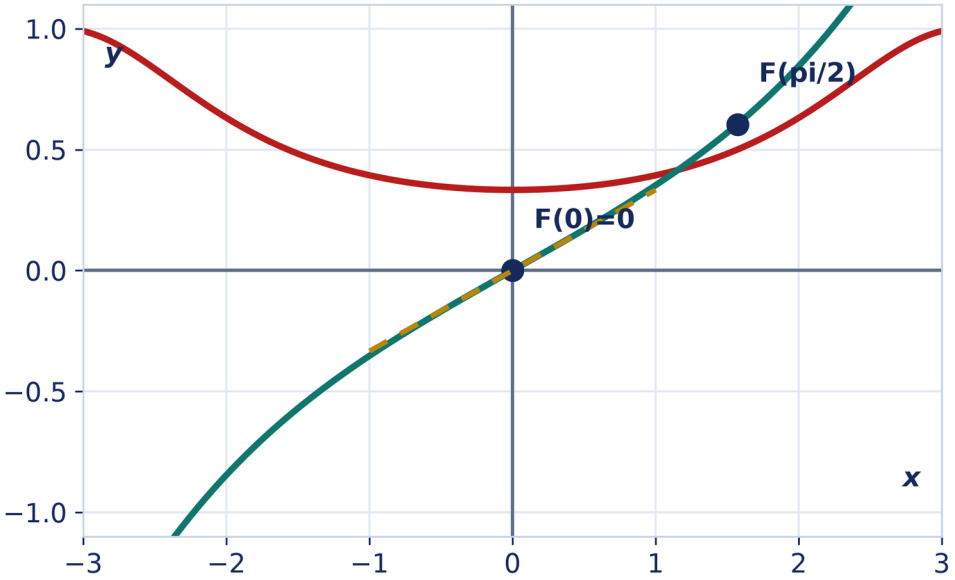


READ THE FIGURE
A=sin^2(x)/2, B=-cos^2(x)/2 and C=-cos(2x)/4. Their vertical differences are A-B=1/2 and A-C=1/4, so all three derivatives equal sin x cos x.

The curves are not approximations of one another: their vertical separations are exactly constant. This is why an answer can look different from a textbook answer and still be completely correct. Differentiate first; then compare constants if necessary.

Lab 6 — Half-angle antiderivatives have branches

The half-angle formula differentiates to 1/(2+cos x) on (-pi,pi)



READ THE FIGURE
Red is f=1/(2+cos x); teal is the half-angle antiderivative F. The gold tangent at zero has slope f(0)=1/3. At pi/2 the slope of F is f(pi/2)=1/2, not the displayed height.

On $-\pi < x < \pi$, $t = \tan(x/2)$ is continuous and strictly increasing, and differentiating the plotted F returns the positive red integrand. At $x = \pm\pi$, the substitution itself fails. A continuous antiderivative on a larger interval must be constructed branch by branch with matching constants.

Extension I — Euler Substitution for Radicals

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 01 / 05

Turn a radical into a rational variable

For $s=\sqrt{x^2+a^2}$, introduce

$$t=x+s>0.$$

Then $s=t-x$ and $s^2=x^2+a^2$, so

$$x=\frac{t^2-a^2}{2t},\qquad s=\frac{t^2+a^2}{2t},\qquad \frac{dx}{s}=\frac{dt}{t}.$$

This is an Euler substitution. It proves the standard form rather than merely quoting it:

$$\int\frac{dx}{\sqrt{x^2+a^2}}=\ln t+C=\ln\left(x+\sqrt{x^2+a^2}\right)+C.$$

The absolute value is unnecessary here because $x+\sqrt{x^2+a^2}>0$ for every real x.

Euler's substitution removes the radical and preserves dx/s

Quantity	Rational form in t	Consequence
t	x+sqrt(x^2+a^2)	t is positive
x	(t^2-a^2)/(2t)	x is rational in t
s	(t^2+a^2)/(2t)	radical is rationalised
dx/s	dt/t	integral becomes logarithmic

READ THE FIGURE

The identity dx/s=dt/t is the key simplification. It follows by differentiating x(t) and comparing with s(t), not by treating dx as a separate number.

The same idea works on a branch of a difference of squares

For $x>a>0$, put $t=x+\sqrt{x^2-a^2}>0$. Now

$$x=\frac{t^2+a^2}{2t},\qquad \sqrt{x^2-a^2}=\frac{t^2-a^2}{2t},$$

and again $dx/s=dt/t$. Thus

$$\int\frac{dx}{\sqrt{x^2-a^2}}=\ln t+C.$$

On $x<-a$, the compact answer uses $\ln|x+\sqrt{x^2-a^2}|$ and is checked separately on that interval.

WORKED EXAMPLE

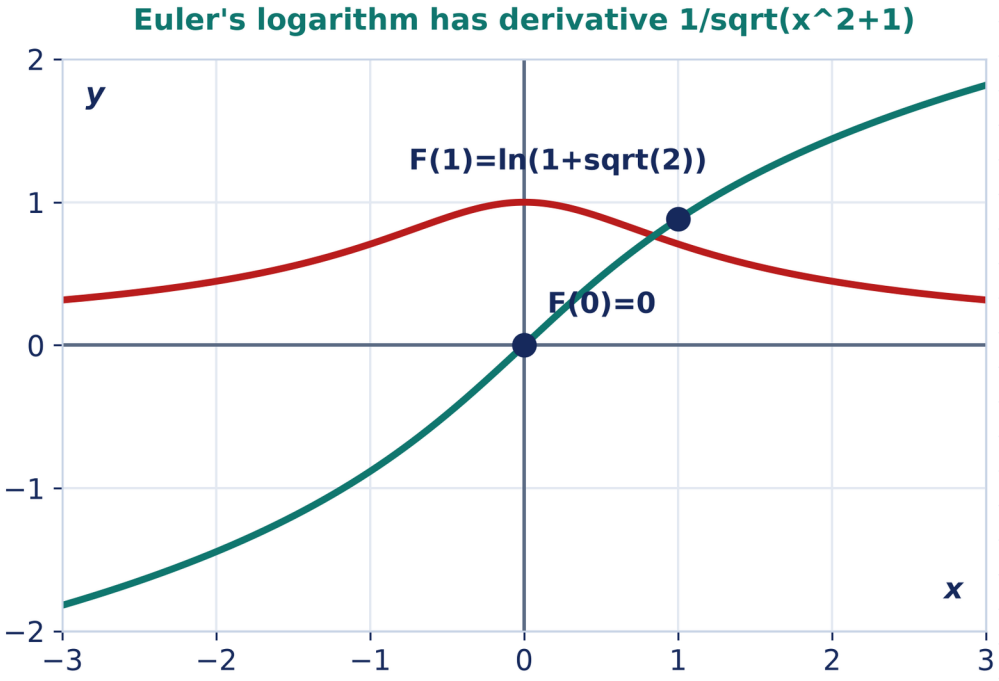
An initial value fixes the logarithmic constant

If $F'(x) = 1/\sqrt{x^2+4}$ and $F(0) = 0$, then

$$F(x) = \ln\left(x + \sqrt{x^2+4}\right) + C.$$

Since the value at zero is $\ln 2 + C$, we need $C = -\ln 2$:

$$F(x) = \ln\left(\frac{x+\sqrt{x^2+4}}{2}\right).$$



READ THE FIGURE

Red is f=1/sqrt(x^2+1), and teal is F=ln(x+sqrt(x^2+1)). F is odd, passes through zero and has positive slope f everywhere.

LESSON 02 / 05

Lower the difficulty one exponent at a time

A reduction formula expresses an integral with exponent n in terms of a smaller exponent. It is a recurrence of antiderivatives, with a constant fixed only at the end.

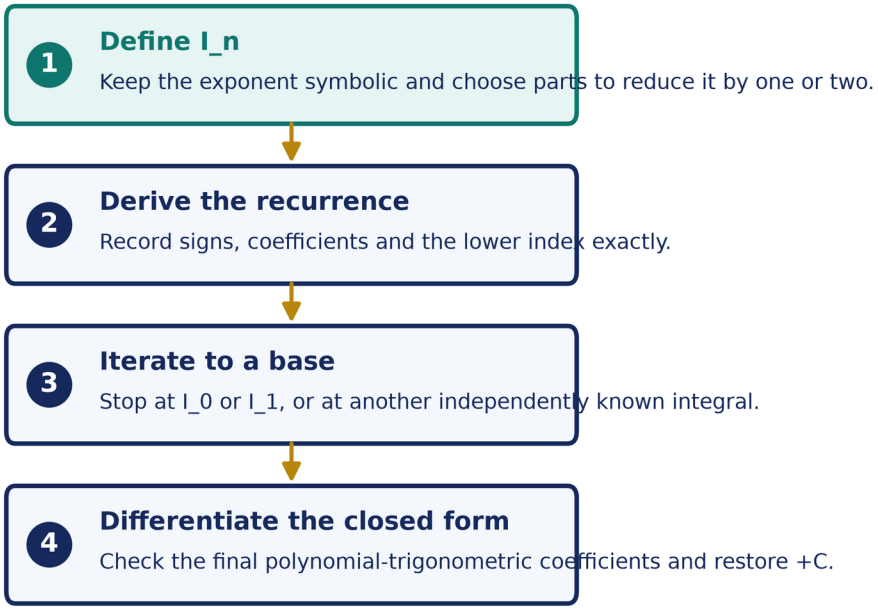
For $I_n = \int x^n e^x dx$, integration by parts gives

$$I_n = x^n e^x - nI_{n-1}, \qquad I_0 = e^x + C.$$

Thus $I_1 = e^x(x-1)$ and $I_2 = e^x(x^2 - 2x + 2)$, followed by

$$I_3 = e^x(x^3 - 3x^2 + 6x - 6) + C.$$

A reduction recurrence terminates at a known base case



READ THE FIGURE
A recurrence is not just a computational shortcut: it also proves the antiderivative's structural form by induction on the exponent.

Logarithmic powers follow the same pattern

For $x > 0$, let $J_n = \int (\ln x)^n dx$. With $u = (\ln x)^n$ and $dv = dx$,

$$J_n = x(\ln x)^n - nJ_{n-1}, \qquad J_0 = x + C.$$

The first three closed forms are

$$J_1 = x\ln x - x, \qquad J_2 = x(\ln x)^2 - 2x\ln x + 2x,$$

$$J_3 = x(\ln x)^3 - 3x(\ln x)^2 + 6x\ln x - 6x,$$

all up to the same final additive constant.

Trigonometric powers step down by two

For $n \geq 2$, let $S_n = \int \sin^n x\,dx$. By parts,

$$S_n = -\tfrac{1}{n}\sin^{n-1} x\cos x + \tfrac{n-1}{n}S_{n-2},$$

with $S_0 = x + C$ and $S_1 = -\cos x + C$. In particular,

$$S_6 = -\tfrac{1}{6}\sin^5 x\cos x - \tfrac{5}{24}\sin^3 x\cos x - \tfrac{5}{16}\sin x\cos x + \tfrac{5}{16}x + C.$$

Differentiating this expression returns $\sin^6 x$. The source's Wallis questions S15, S19, Q30 and Q43 apply limits to this same recurrence in Chapter 21.

COMMON TRAP

The recurrence's constant belongs to the completed answer

The displayed recurrence is an equality of chosen antiderivatives. When iterating, keep a consistent choice and add the arbitrary constant at the end. Do not allow unrelated constants at each stage to masquerade as extra terms.

Extension III — Parameter Differentiation

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 03 / 05

A parameter can manufacture a logarithm

For $x>0$ and $a\neq -1$, start with

$$H(a,x)=\int x^a dx=\frac{x^{a+1}}{a+1}+C.$$

Holding x fixed and differentiating with respect to a brings down $\ln x$:

$$\frac{\partial H}{\partial a}=\int x^a \ln x\,dx=\frac{x^{a+1}((a+1)\ln x-1)}{(a+1)^2}+C.$$

At $a=0$ this reproduces $\int \ln x\,dx=x\ln x-x$. At $a=1$ it gives

$$\int x\ln x\,dx=\frac{x^2}{2}\ln x-\frac{x^2}{4}+C,$$

which is the antiderivative used in the source's S14 and Q28.

Parameter differentiation creates a logarithmic hierarchy

Starting object	Operation in a	New integrand
x^(a+1)/(a+1)	first derivative	x^a ln x
x^(a+1)/(a+1)	second derivative	x^a (ln x)^2
a=0 after first	specialise	ln x
a=1 after first	specialise	x ln x

READ THE FIGURE

Differentiate first with a general, then substitute the parameter value. Setting a=0 before differentiating would erase the parameter being used.

A second differentiation gives squared logarithms

Differentiating once more,

$$\int x^a(\ln x)^2dx=x^{a+1}\left(\frac{(\ln x)^2}{a+1}-\frac{2\ln x}{(a+1)^2}+\frac{2}{(a+1)^3}\right)+C.$$

At $a=0$,

$$\int (\ln x)^2dx=x(\ln x)^2-2x\ln x+2x+C,$$

in agreement with the reduction formula J_2 in Extension II. This is the indefinite form behind P14 and Q29; their endpoint limits belong to Chapter 21.

WORKED EXAMPLE

Cross-check two methods at a=2

Parameter differentiation gives

$$\int x^2 \ln x \, dx = x^3\left(\frac{\ln x}{3} - \frac{1}{9}\right) + C.$$

Direct by parts with $u = \ln x$ and $dv = x^2 dx$ gives exactly the same expression. Their agreement checks both the parameter derivative and the by-parts coefficient.

COMMON TRAP

State the domain and the order of operations

Here $x > 0$ makes $x^a = e^{a \ln x}$ unambiguous. Differentiate with respect to a while x is fixed, and only then specialise a. The method generates an antiderivative; any later definite limit requires its own convergence check.

LESSON 04 / 05

Some antiderivatives require new named functions

An elementary function is built from algebraic, exponential, logarithmic, trigonometric and inverse-trigonometric operations. Liouville's theory shows that several innocent-looking integrands have no antiderivative within that class.

This does **not** say that no antiderivative exists. It says that we must enlarge the function library.

Special functions can be introduced by derivative and normalisation

Integrand	Named antiderivative	Defining derivative
exp(-x^2)	(sqrt(pi)/2) erf(x)	erf'(x)=2 exp(-x^2)/sqrt(pi)
sin x / x	Si(x)	Si'(x)=sin x/x
1 / ln x	li(x)	li'(x)=1/ln x
exp(x) / x	Ei(x)	Ei'(x)=exp(x)/x

READ THE FIGURE
Together with erf(0)=0 and Si(0)=0, the first two derivatives determine smooth special functions. The logarithmic and exponential integrals require domain branches around their singularities.

For example,

$$\int e^{-x^2}dx = \frac{\sqrt{\pi}}{2}\operatorname{erf}(x) + C, \qquad \operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}}e^{-x^2}.$$

Similarly, after assigning the continuous value 1 to $\sin x/x$ at $x = 0$,

$$\int \frac{\sin x}{x}dx = \operatorname{Si}(x) + C, \qquad \operatorname{Si}(0) = 0.$$

These statements are verified by the same principle as every other antiderivative: differentiate the named function.

The frontier is about representation, not computational failure

The source's Gaussian S16/Q32/Q46 and Dirichlet Q37 questions concern **definite** values. Chapter 21 will show that some of those values are exact even though no elementary antiderivative is available. A definite symmetry or double-integral method can succeed where an elementary formula for $F(x)$ does not exist.

COMMON TRAP

Never write 'the integral does not exist' merely because the usual methods fail

The precise statement is "there is no elementary antiderivative." Numerical approximation, a special function, a convergent series or an exact definite-integral technique may still be available. Conversely, quoting a special-function name without its derivative and domain is not a verification.

LESSON 05 / 05

One complex exponential integrates sine and cosine together

Using $i^2 = -1$ and $e^{ix} = \cos x + i\sin x$,

$$\int e^{(a+ib)x}dx = \frac{e^{(a+ib)x}}{a+ib} + C, \qquad a+ib \neq 0.$$

Taking real and imaginary parts gives the general formulas

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C,$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + C.$$

At $a = b = 1$, the sine formula becomes Q18's $\frac{1}{2}e^x(\sin x - \cos x)$.

An ansatz gives the same result without complex arithmetic

Try

$$F = e^{ax}(A\sin bx + B\cos bx).$$

Then

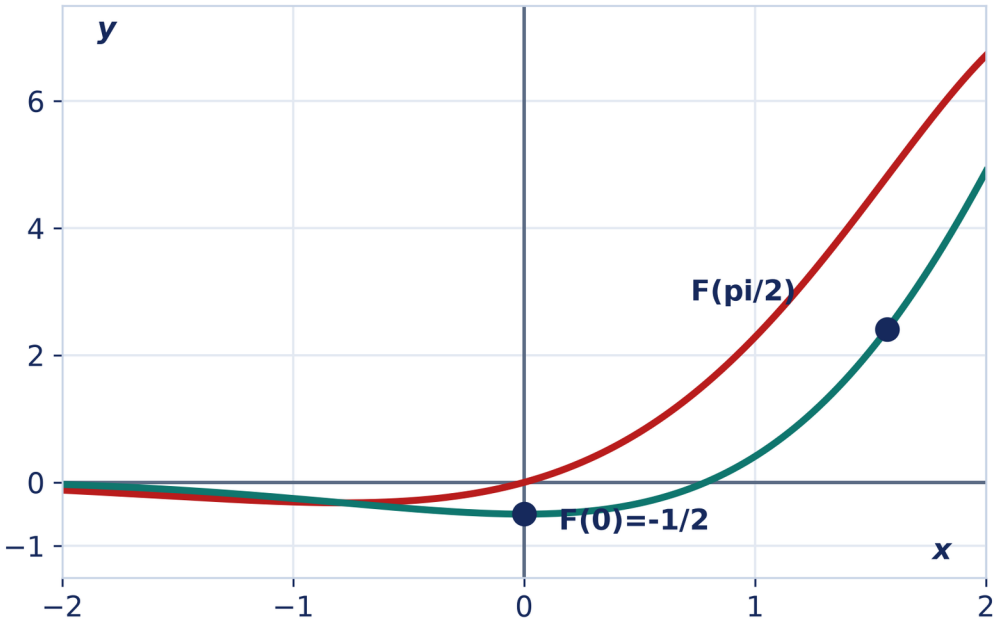
$$F' = e^{ax}((aA - bB)\sin bx + (bA + aB)\cos bx).$$

To make $F' = e^{ax}\sin bx$, solve

$$aA - bB = 1, \qquad bA + aB = 0,$$

which gives $A = a/(a^2+b^2)$ and $B = -b/(a^2+b^2)$. The cosine case only changes the right-hand sides of these two linear equations.

The cyclic antiderivative has slope e^x sin x



READ THE FIGURE

Red is f=e^x sin x; teal is F=e^x(sin x-cos x)/2. The slope of teal at zero is f(0)=0, while at pi/2 it is f(pi/2)=e^(pi/2), about 4.8105.

WORKED EXAMPLE

The same ansatz handles a damped cosine

With $a=-2$ and $b=3$,

$$\int e^{-2x}\cos 3x\,dx=\frac{e^{-2x}(-2\cos 3x+3\sin 3x)}{13}+C.$$

Differentiation restores both exponential damping and the frequency 3; the denominator is $(-2)^2+3^2=13$.

COMMON TRAP

Do not mix the sine and cosine coefficient rows

The sine antiderivative has $a\sin bx-b\cos bx$; the cosine antiderivative has $a\cos bx+b\sin bx$. A quick differentiation at one general parameter set is safer than relying on visual similarity between the two formulas.